We know that principal value of $\sin\theta$ lies in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Since, $\sin\theta = -\frac{\sqrt{3}}{2} < 0$, the principal value of $\sin\theta$ lies in the IV quadrant.

$\sin\theta = -\frac{\sqrt{3}}{2} = -\sin\left(\frac{\pi}{3}\right) = \sin\left(-\frac{\pi}{3}\right)$.

Hence, $\theta = -\frac{\pi}{3}$ is the principal solution.

(iii) $\operatorname{cosec}\theta = -2$

$\operatorname{cosec}\theta = -2 \Rightarrow \frac{1}{\sin\theta} = -2 \Rightarrow \sin\theta = -\frac{1}{2}$

Since $\sin\theta = -\frac{1}{2} < 0$, the principal value of $\sin\theta$ lies in the IV quadrant.

$\sin\theta = -\frac{1}{2} = -\sin\frac{\pi}{6} = \sin\left(-\frac{\pi}{6}\right)$

Thus, $\theta = -\frac{\pi}{6}$ is the principal solution.

(iv) $\cos\theta = \frac{1}{2}$

Principal value of $\cos\theta$ lies in the I and II quadrant.

Since $\cos\theta = \frac{1}{2} > 0$, the principal value of $\cos\theta$ lies in the interval $\left[0, \frac{\pi}{2}\right]$.

$\cos\theta = \frac{1}{2} = \cos\frac{\pi}{3}$

Thus, $\theta = \frac{\pi}{3}$ is the principal solution.

Example 3.43 Find the general solution of $\sin\theta = -\frac{\sqrt{3}}{2}$

Solution:
The general solution of $\sin\theta = \sin\alpha, \alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, is $\theta = n\pi + (-1)^n \alpha, n \in \mathbb{Z}$

$$\sin\theta = -\frac{\sqrt{3}}{2} = \sin\left(-\frac{\pi}{3}\right),$$

Thus the general solution is

$$\theta = n\pi + (-1)^n\left(-\frac{\pi}{3}\right) = n\pi + (-1)^{n+1}\left(\frac{\pi}{3}\right); n \in \mathbb{Z} \quad \text{... (i)}$$

In arriving at the above general solution, we took the principal value as $-\left(\frac{\pi}{3}\right)$ with our convention that principal value is the numerically smallest one in the interval $[-\pi, \pi]$. Now through this example, we shall justify that the principal value may also be taken in $[0, 2\pi]$,